Tushar Bhat

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Education

Purdue University West Lafayette, IN
Bachelor of Science Aerospace Engineering, Minor: Computer Science Fall 2024 – Spring 2028
Relevant Coursework: Turbomachinery, Fluid Mechanics, Aerodynamics, Structures, Signal Analysis, Propulsion Foundations, Intro to CFD, Thermo

Skills

- MATLAB, ANSYS Workbench (CFD, FEA), Siemens NX, CAD/CAM, XFLR5, Teamcenter, GD&T, Additive Manufacturing, CNC Machining, Soldering, MS Office, Agile Principles, Python, GitHub, Data Analysis, Basic ML, Signal Processing

Work Experience

HCLTech – *Systems R&D Intern* Redmond, WA
Boeing Contracting – June 2026 – August 2026

- Developed an RSSI fingerprinting localization model in MATLAB for a GPS-denied metallic cargo environment, using KNN to match signal measurements against reference fingerprints and achieving a 1.05 m median localization error across five simulated test positions.
  - Built a custom channel model capturing path loss and multipath effects, using 3D ray tracing to generate synthetic RSSI data and reference fingerprints for validating and stress-testing localization accuracy without access to physical aircraft testing.
- Collaborated directly with a senior engineering manager to develop the localization approach and anchor geometry, jointly shaping the system design and tracking strategy from initial concept through working simulation.

Purdue Data Mine & The Aerospace Corporation – *RF Signal Processing Analyst* West Lafayette, IN
GNSS Interference Signal Classification August 2025 – May 2026

- Partnered with The Aerospace Corporation to develop an ML pipeline for automated GNSS jamming detection and classification across a 20+ GB RF dataset; presented methodology and results to industry partners and professors at the Purdue Data Mine Symposium.
- Designed a custom 2D ResNet with stochastic depth in PyTorch Lightning to classify power spectral density matrices into nine interference classes, processing signal matrices as single-channel images through a residual stem and three residual stages spanning 32–256 channels.
  - Addressed severe data imbalance and label noise through synthetic data generation, inverse class weighting, and real-time Gaussian noise augmentation, improving model robustness on underrepresented interference classes.
  - Achieved 68% model accuracy and a 0.6 F1 score, contributing to an 11-model ensemble that reached 76.91% test accuracy

Purdue Undergraduate Rocket Propulsion Labs – *Senior Design Engineer* West Lafayette, IN
Testbed Project – October 2025 - Present

- Collaborated with a six-person simulation team responsible for thermal-fluid analysis of a 500 lbf LOx–RP-1 regeneratively cooled rocket engine, setting analysis priorities and coordinating boundary conditions and load cases with design, manufacturing, and test subteams to support CDR and hotfire readiness.
- Retained hands-on ownership of the highest-risk regen channel analyses, executing thermal-fluid CFD in ANSYS Fluent/CFX with sequential structural coupling to evaluate channel wall temperatures and coolant side pressures ahead of high-pressure testing.
  - Performed structural FEA in ANSYS Mechanical to establish operating limits and safety margins for pressure vessels and engine hardware, evaluating factors of safety against anticipated hotfire operating conditions.
- Established project-wide CAD and geometry preparation standards in Siemens NX and ANSYS Discovery after defective geometry compromised the initial CDR simulations, eliminating the geometry errors responsible for incomplete and inaccurate analysis results.
- Currently developing the club's first resonance igniter, owning concept development from project initiation through Phase 1 Design Review, including requirements definition, propellant selection, and preliminary design.

Purdue Propulsion and Energy Lab – *Undergraduate Research Assistant* West Lafayette, IN
Autonomous UAV Hydrogen Refueling – August 2025 - May 2026

- Researched a mechanical docking and fluid-transfer system design for autonomous UAV hydrogen refueling, addressing alignment, clamping, leak prevention, and secure connection requirements.
- Conducted trade studies and built decision matrices to compare docking concepts and guide design selection.
- Consulted vendors and machinists on structural material options and alternative docking designs to inform material and mechanism selection and manufacturing.

Projects

Torch Igniter August 2025 – Present

- Independently designed and CNC machined a Methane-GOX torch igniter along with a Combustion Simulation benchmarked against a NASA CEA ideal baseline, achieving 84% efficiency and confirming a 1,700 K exit temperature sufficient to ignite the main chamber.

Autonomous Fixed Wing UAV May 2025 – August 2025

- Designing and prototyping using XFLR5 and NX, a fixed wing autonomous UAV with a wingspan of 2.65m capable of achieving a speeds of 30mph
- Integrated and bench tested the complete avionics stack – Flight Controller, Raspberry Pi, GPS, Motors, Servos, etc. – validating system functionality

Axisymmetric Ramjet Design:

- Designing a Mach 3.2 axisymmetric mixed-compression ramjet for testing; leading inlet design and evaluating shock geometry and pressure recovery using NASA's SUPIN software.

Additional Experiences

Eastlake High School – Link Crew August 2022 – June 2024

- Acted yearlong (x2) as a guide to 6 freshmen by directing them at school events, assisting with club sign-ups, facilitating workshops, and leading campus tours.

Museum of Flight – Pavillion Ambassador August 2022 – June 2024

- Supervised and regulated aircraft displays. Educated visitors on planes' history and technical abilities ranging from WWII bombers to 787 Dreamliners.